



myoRESEARCH 3.18+

Software Release Notes

NORAXON®



DISCLAIMER:

We recommend backing up your existing installation version & database before installing the latest version.

Please be advised that the update should not be applied to existing versions used in ongoing research studies as some calculations have changed and may present different results than previous versions.

If you would like to utilize the new features for additional projects, we recommend installing the new version in a separate directory folder by changing the default installation path to a new folder, like c:\program files\noraxon\mr3.18.

For assistance, please contact Noraxon Technical Support at support@noraxon.com.

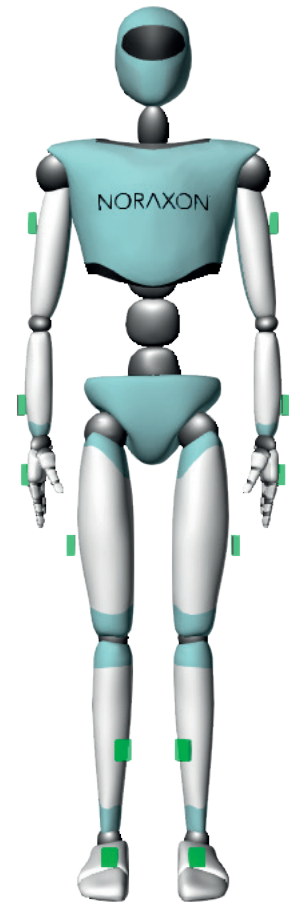
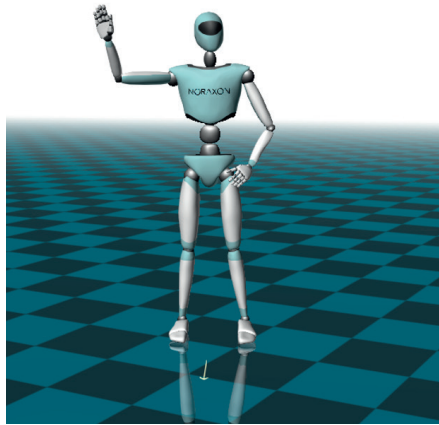
Ultium Battery Life Indicator

Battery life indicators for all Ultium sensors are provided in MR3 allowing users to monitor battery life in real-time and receive alerts for low charge levels.

Ultium Motion			Ref	0.920		-43.2	
	Normal	60121 Foot LT	Uppe...	0.903	-0.017	-43.2	-0.0
	Normal	60193 Thigh LT	Head	1.036	0.115	-43.3	-0.1
	Normal	6013c Upper spine	Uppe...	0.926	0.005	-42.8	0.4
	Normal	60158 Head	Forea...	0.934	0.014	-43.6	-0.4
	Normal	60154 Upper arm RT	Thigh...	0.919	-0.002	-43.1	0.1
	Normal	60137 Upper arm LT	Foot LT	0.899	-0.021	-44.5	-1.3
	Normal	6012e Forearm LT	Uppe...	0.883	-0.037	-41.7	1.5
Ultium EMG 2							
	Normal	20172 EMG 1					
	Normal	20134 EMG 2					
	Normal	20492 EMG 3					
	Normal	20627 EMG 4					
	Normal	20116 EMG 5					
	Normal	20115 EMG 6					
	Normal	20114 EMG 7					
	Normal	20a09 EMG 8					
				Battery: Normal			
				Calibrate Baselines			
				Screen Layout			
				EMG Baseline Check			
			Calibrate				

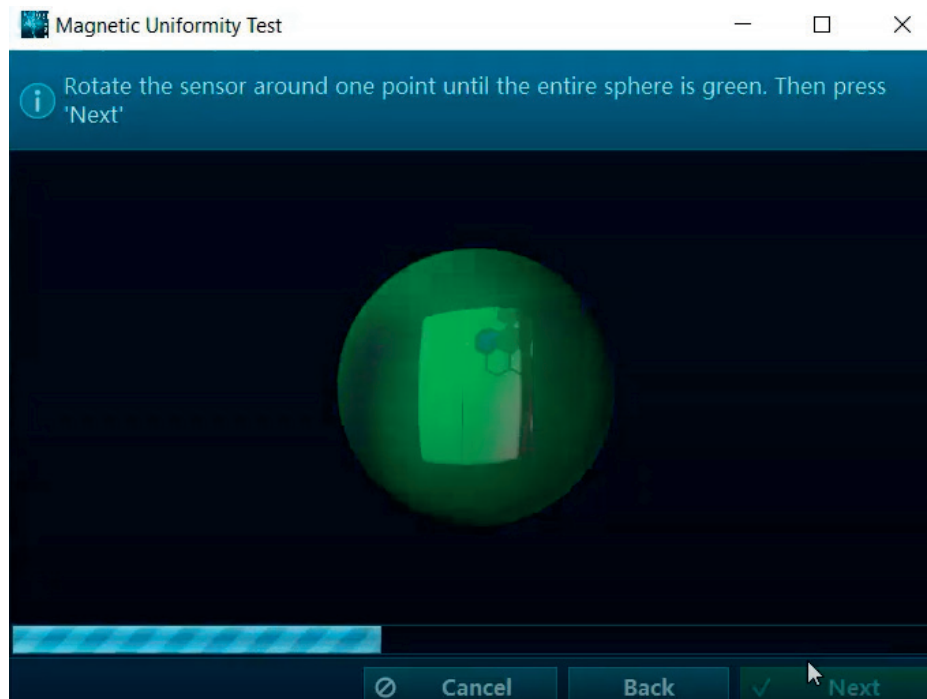
Meet Movi - Our New Avatar!

An alternative avatar is now available for users who do not wish to use the skeletal avatar for their applications.



Sensor Demagnetization Tool

A tool has been added to MR3 allowing for the field calibration of the magnetometer within an Ultium Motion sensor.

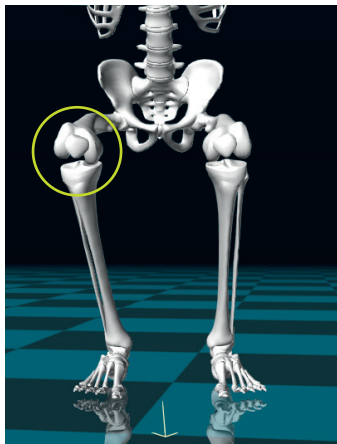


NOTE: This tool is only accessible by Noraxon support personnel and should only be used if it is determined that a sensor's magnetometer has become distorted.

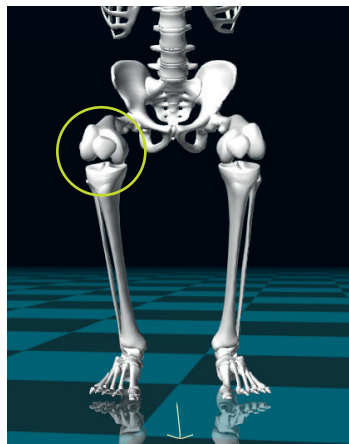
Animation Optimizations

- A natural adduction of the femur in neutral standing has been applied to the model.
- Avatar animation has been improved to better handle femur roll artifact.

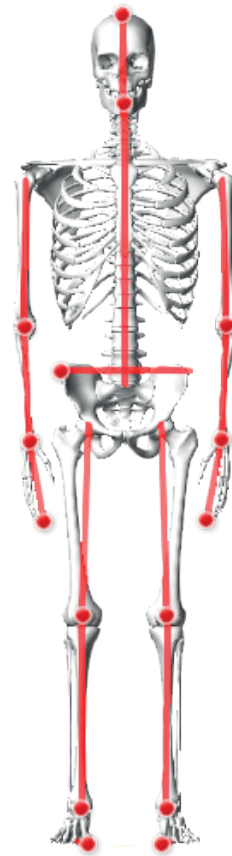
Note: Animation updates do not affect data outputs.



Previous

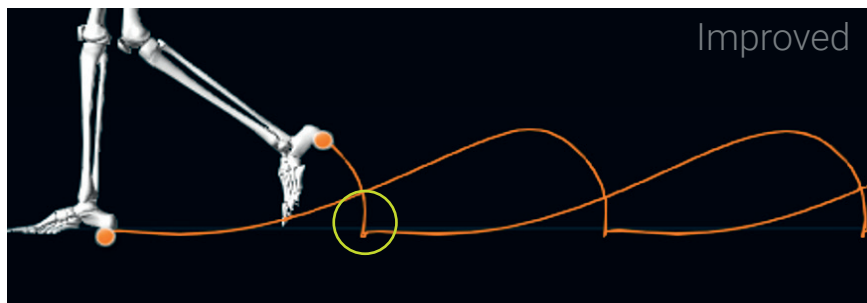


Improved

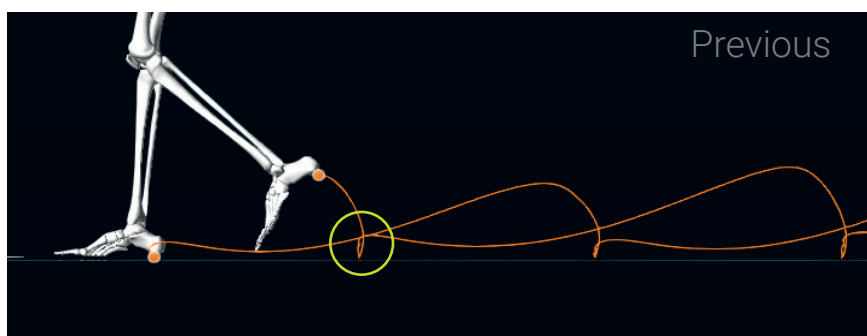


Calibration Offset Adjustments

A 5° forward tilt has been applied to the thigh & shank segments to produce a more natural and balanced center of mass position during the default calibration position. This results in a reduction of artifacts in gait contacts previously due to the unnatural vertical positioning of the lower limbs underneath the pelvis.



Improved



Previous



NOTE: Ankle Dorsiflexion, Hip Flexion, as well as the Thigh and Shank segment angles now present with a 5° offset in the default calibration position.

Expanded Single Segment Angles

Single segment angles have been added for all the anatomical body segments.

NOTE: Angle labels have been updated for the previous single segment angles (Pelvis and Foot).

Sensors	Segment Angles	Joint Angles	Orientations	Advanced
Select angles to be shown. If you do not see your angles, select the necessary sensors.				
	Side	Name		
☒		Head Tilt Fwd		
☒	LT	Head Tilt Lat		
☒	RT	Head Tilt Lat		
☒	RT	Head Rotation Ext		
☒	LT	Head Rotation Ext		
☒	LT	Upper Arm Tilt Fwd		
☒	RT	Upper Arm Tilt Fwd		
☒	LT	Upper Arm Tilt Med		
☒	RT	Upper Arm Tilt Med		
☒	LT	Upper Arm Rotation Ext		
☒	RT	Upper Arm Rotation Ext		
☒	RT	Forearm Tilt Fwd		
☒	RT	Forearm Tilt Med		
☒	RT	Forearm Rotation Ext		
☒	RT	Hand Tilt Fwd		
☒	RT	Hand Tilt Med		
☒	RT	Hand Rotation Ext		
☒		Upper Spine Tilt Fwd		
☒	LT	Upper Spine Tilt Lat		
☒	RT	Upper Spine Tilt Lat		
☒	LT	Upper Spine Rotation		
☒	RT	Upper Spine Rotation		
☒		Lower Spine Tilt Fwd		
☒	LT	Lower Spine Tilt Lat		

☒	LT	Lower Spine Rotation
☒	RT	Lower Spine Rotation
☒		Pelvic Tilt Fwd
☒	LT	Pelvic Tilt Lat
☒	RT	Pelvic Tilt Lat
☒	LT	Pelvic Rotation Ext
☒	RT	Pelvic Rotation Ext
☒	LT	Thigh Tilt Fwd
☒	RT	Thigh Tilt Fwd
☒	LT	Thigh Tilt Med
☒	RT	Thigh Tilt Med
☒	LT	Thigh Rotation Ext
☒	RT	Thigh Rotation Ext
☒	LT	Shank Tilt Fwd
☒	RT	Shank Tilt Fwd
☒	LT	Shank Tilt Med
☒	RT	Shank Tilt Med
☒	LT	Shank Rotation Ext
☒	RT	Shank Rotation Ext
☒	LT	Foot Tilt Fwd
☒	RT	Foot Tilt Fwd
☒	LT	Foot Tilt Med
☒	RT	Foot Tilt Med
☒	LT	Foot Rotation Ext
☒	RT	Foot Rotation Ext

Lean Forward Calibration

An additional functional calibration method has been added for a quicker and easier alternative if the physical environment is not suitable for the Walking calibration method.

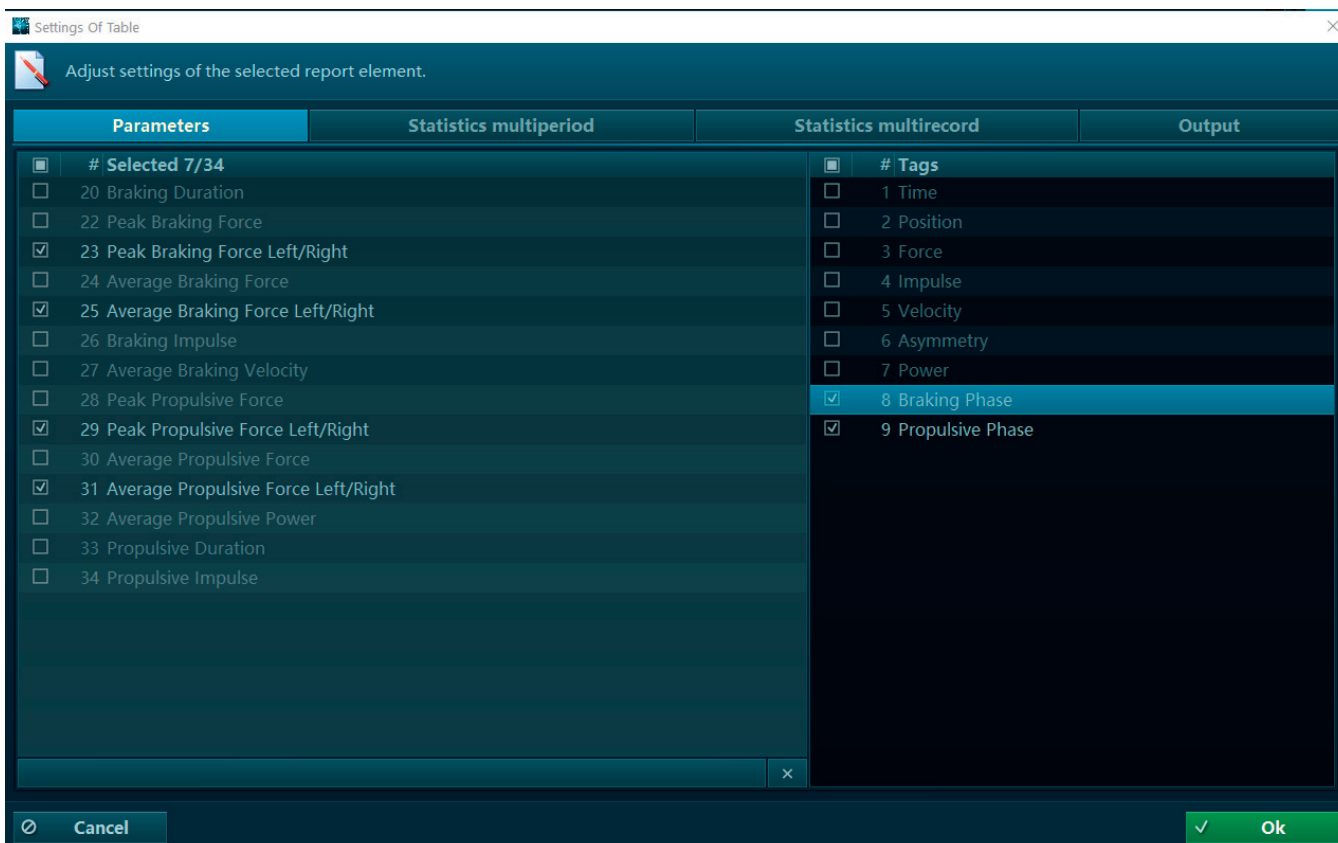


← Standing Pose	← A-frame Pose	← Forward Lean	← Standing Pose
 Take the calibration pose and press 'Capture'.	 Abduct your arms by ≈30 deg and hold, press 'Capture'.	 Lean forward and hold. Press 'Capture'.	 Take the calibration pose again. Press 'Capture'.
			

Lean Forward Calibration Sequence

Jump Report Parameter Filters

Parameter tags are used to provide a simple filtering scheme for the 35+ vertical jump parameters available in the jump report.



More Jump Parameters

Additional jump parameters have been added for a total of 40 different vertical jump parameters.

- Average Propulsive Velocity, m/s
- Force At Zero Velocity, N
- Average Braking Power, Watt
- Max Rate of Power Development, Watt/s
- Countermovement Depth, cm

Phase Definition for Jump Loops

The jump loop report element can now be restricted to the loading or landing subphases and the force inputs can be side separated.

NOTE: A dual force plate setup is required for side separated force data.

